



NORTH SYDNEY BOYS HIGH SCHOOL

2022 YEAR 11 ASSESSMENT TASK 3

Mathematics Extension 1

General Instructions

- Working time – **70 minutes** + 5 mins reading time
- Write in this booklet
- Write using blue or black **pen**
- Board approved calculators may be used
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class Teacher: Please shade the circle.

- ☐ Mr Berry
- ☐ Ms Fu
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Moss
- ☐ Mr Umakanthan

STUDENT NAME:

(To be used by the exam markers only.)

Section	A	B	C	D	E	F	Tot	Tot
Mark	$\overline{12}$	$\overline{12}$	$\overline{9}$	$\overline{10}$	$\overline{12}$	$\overline{8}$	$\overline{63}$	$\overline{100}$

Section A (12 marks)

1. Let α, β , and γ be the roots of $x^3 - 4x + 1 = 0$.

1

Circle the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$:

(A) -4

(B) -1

(C) 1

(D) 4

2. The polynomial $P(x) = x^3 - x^2 - 5x - 3$ has a double root at $x = \alpha$.

What is the value of α ?

1

(A) $\alpha = -\frac{5}{3}$

(B) $\alpha = -1$

(C) $\alpha = 1$

(D) $\alpha = \frac{5}{3}$

3. The polynomial $P(x)$ is given by $P(x) = 2x^3 + 7x^2 - 46x + 21$.

(i) Show that $P(x)$ is divisible by $(x - 3)$. 1

(ii) Hence, fully factorise $P(x)$. 3

4. The polynomial $P(x) = x^3 + 2x^2 + ax + b$ has a factor of $x + 2$ and when divided by $x - 2$ there is a remainder of 12. Find the values of a and b .

3

5. The polynomial equation $4x^3 - 12x^2 + 5x + 6 = 0$ has roots α, β , and γ . It is known that one of the roots is the sum of the other two. Find α, β , and γ .

3

Section B (12 marks)

1. What is the value of $\cos^{-1}(-\frac{1}{2})$? 1

(A) $\frac{\pi}{3}$

(B) $-\frac{\pi}{3}$

(C) $\frac{2\pi}{3}$

(D) $\frac{4\pi}{3}$

2. Find the **exact** values of the following:

(i) $\tan^{-1}(-\sqrt{3})$ 1

(ii) $\sin^{-1}(\sin \frac{7\pi}{5})$ 1

(iii) $\sin(\cos^{-1} \frac{-2}{3})$ 2

(iv) $\tan(2 \sin^{-1} \frac{3}{4})$ 2

3.

(i) State the domain and range of the function $f(x) = 3 \sin^{-1} \left(\frac{x}{2} \right)$ 2

(ii) Sketch the graph $y = f(x)$ 2

4.

Which diagram represents the domain of the function $f(x) = \cos^{-1}\left(\frac{3}{x}\right)$?

1

(A)



(B)



(C)



(D)



Section C (9 marks)

1 Find the coefficient of x^2 in the expansion of $(x^2 + \frac{2}{x})^7$

2

2. Consider the expansion of $(1 + 2x)^n$

(i) Write down an expression for the coefficient of the term in x^4

2

(ii) The ratio of the coefficient of the term in x^4 to that of the term in x^6 is 5 : 8
Find the value of n .

3

3. Consider the binomial expansion $(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$

Show that $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} = n2^{n-1}$

2

Section D (10 marks)

1. What is the value of $\sin 2\theta$ given that $\sin \theta = \frac{\sqrt{3}}{4}$ and θ is obtuse?

1

(A) $\frac{-\sqrt{39}}{8}$

(B) $\frac{-\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{\sqrt{39}}{8}$

2. Using the t -substitutions, or otherwise, prove the identity

3

$$\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$$

3. Using an appropriate products-to-sums relationship, or otherwise, solve the equation

$$\sin 5x + \sin x = 0 \quad \text{in the domain } 0 \leq x \leq \frac{\pi}{2}.$$

2

4. The quadratic equation $ax^2 + bx + c = 0$ has roots $x = \tan \alpha$ and $x = \tan \beta$.

(i) Show that $\tan (\alpha + \beta) = \frac{b}{c-a}$ 2

(ii) Show that $\tan^2 (\alpha - \beta) = \frac{b^2 - 4ac}{(a+c)^2}$ 2

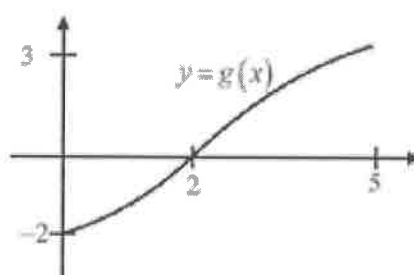
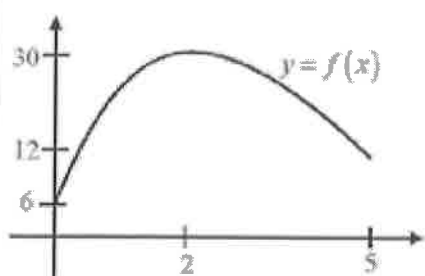
Section E (12 marks)

1. Solve the inequality $\frac{2x}{1-x} \geq x$

3

2. Consider the following sketches of the functions $y = f(x)$ and $y = g(x)$.

1



What is the range of the function $y = \frac{f(x)}{g(x)}$? [Circle the answer]

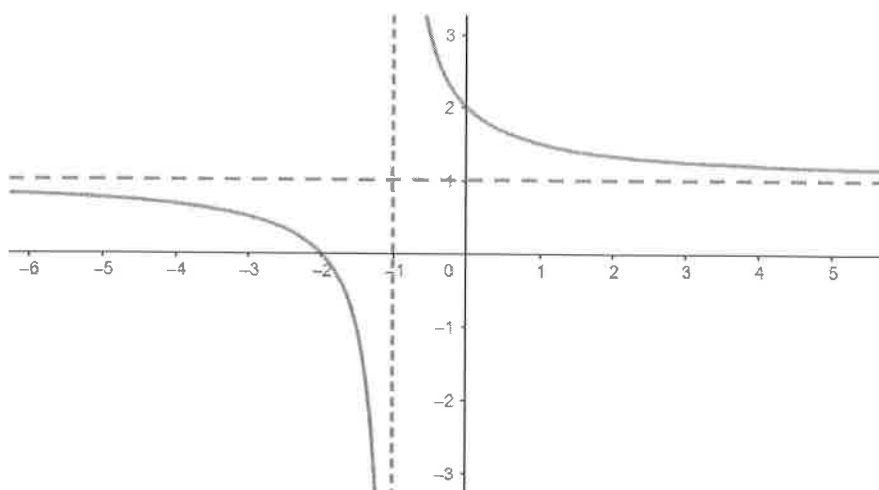
(A) $(-\infty, -3] \cup [4, \infty)$

(B) $[-3, 4]$

(C) $[-15, 2]$

(D) all real y

3. The graph of $y = f(x)$ is shown below:



Draw separate one-third page sketches of the following curves, clearly indicating any important features such as asymptotes, intercepts and turning points.

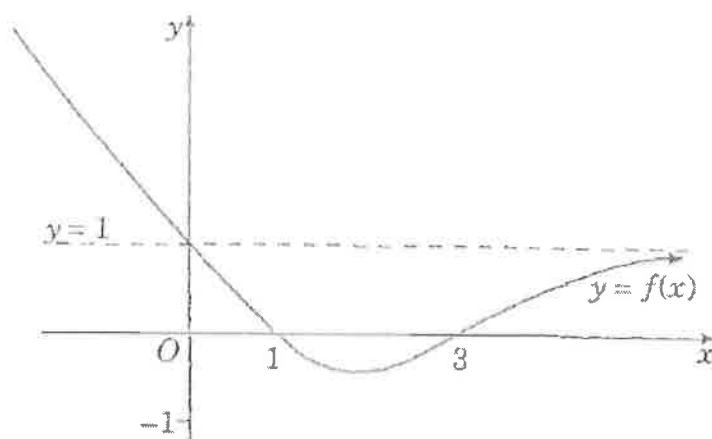
(i) $y = |f(x)|$

2

(ii) $y = f(|x|)$

2

4. The graph shows the graph of the function $y = f(x)$. It has a horizontal asymptote $y = 1$.

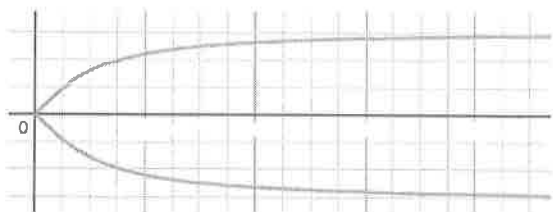


Draw a one-third page graph of $y = \sqrt{f(x)}$ showing any asymptotes and stating its domain and range.

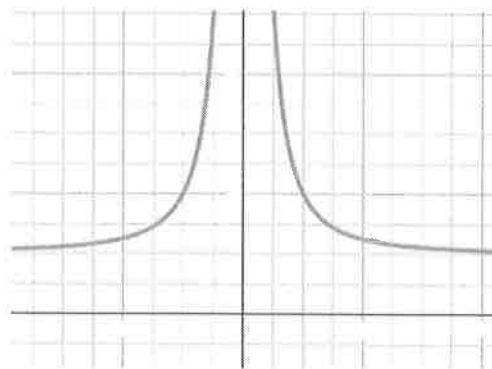
5. Which of the following graphs best represents $|y| = \tan^{-1}(x)$?

1

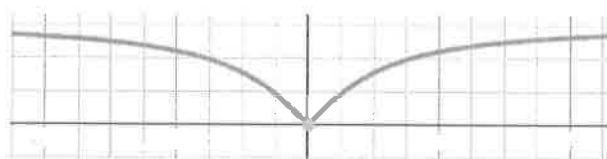
(A)



(B)



(C)



(D)



Section F (8 marks)

1. The surface area of a sphere of radius r metres is decreasing at a rate of 0.8 m^2 per second at the instant when $r = 2.3 \text{ m}$.

Calculate the rate of decrease of the radius of the sphere at this instant.

2

2. A metal statuette is taken from a freezer at -8°C into a room where the air temperature is 22°C .

The rate at which the statuette warms follows Newton's law, that is $\frac{dT}{dt} = -k(T - 22)$ where k is a positive constant, time t is measured in minutes, and temperature T is measured in degrees Celsius.

- (i) Show that $T = 22 - Ae^{-kt}$ is a solution of the equation $\frac{dT}{dt} = -k(T - 22)$, and find the value of A .

2

- (ii) The temperature of the statuette reaches 4°C in 90 minutes. Find the exact value of k .

2

- (iii) Find the temperature of the statuette after another 90 minutes.

1

3. When added to water, 5 grams of a substance dissolves at a rate equal to 10% of the amount of undissolved chemical per hour. If x is the number of grams of *undissolved* chemical after t hours, which equation describes the rate of change of x ? **1**

(A) $\frac{dx}{dt} = -\frac{1}{10}x$

(B) $\frac{dx}{dt} = -\frac{1}{5}x$

(C) $\frac{dx}{dt} = \frac{1}{5}(10 - x)$

(D) $\frac{dx}{dt} = \frac{1}{10}(5 - x)$

END OF EXAM



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Section A (12 marks)

1. Let α, β , and γ be the roots of $x^3 - 4x + 1 = 0$.

1

Circle the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$:

(A) -4

(B) -1

(C) 1

(D) 4

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$= \frac{-4}{-1} = 4 \quad \therefore \textcircled{D} \quad \checkmark$$

2. The polynomial $P(x) = x^3 - x^2 - 5x - 3$ has a double root at $x = \alpha$.

What is the value of α ?

1

(A) $\alpha = -\frac{5}{3}$

(B) $\alpha = -1$

(C) $\alpha = 1$

(D) $\alpha = \frac{5}{3}$

$$P'(x) = 3x^2 - 2x - 5$$

$$= (3x - 5)(x + 1) = 0$$

$$\therefore x = -1 \text{ or } x = \frac{5}{3}$$

$$\text{Since } P(-1) = -1 - 1 + 5 - 3 = 0,$$

and there can only be 1 double root tops,

$$\therefore \text{double root is } -1 \quad \therefore \textcircled{B} \quad \checkmark$$

3. The polynomial $P(x)$ is given by $P(x) = 2x^3 + 7x^2 - 46x + 21$.

(i) Show that $P(x)$ is divisible by $(x - 3)$.

1

$$P(3) = 2(3^3) + 7(3^2) - 46(3) + 21 = 54 - 63 - 138 + 21 = 0 \quad \checkmark$$

(ii) Hence, fully factorise $P(x)$.

$\therefore x-3$ is factor.

3

$$\begin{array}{r} 2x^2 + 13x - 7 \quad \checkmark \\ x-3 \overline{) 2x^3 + 7x^2 - 46x + 21} \\ \underline{2x^3 - 6x^2} \times \\ 13x^2 - 46x \\ \underline{13x^2 - 39x} \\ -7x + 21 \\ \underline{-7x + 21} \\ 0 \end{array}$$

$$\therefore P(x) = (x-3)(2x^2 + 13x - 7)$$

$$P(x) = (x-3)(2x-1)(x+7) \quad \checkmark \checkmark$$

4. The polynomial $P(x) = x^3 + 2x^2 + ax + b$ has a factor of $x + 2$ and when divided by $x - 2$ there is a remainder of 12. Find the values of a and b .

3

$$P(-2) = 0 \quad \therefore (-2)^3 + 2(-2)^2 + a(-2) + b = 0$$
$$-8 + 8 - 2a + b = 0$$

$$\therefore -2a + b = 0 \quad \text{--- (1)}$$

$$P(2) = 12$$

$$\therefore 2^3 + 2(2^2) + 2a + b = 12$$

$$\therefore 2a + b = -4 \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \rightarrow 2b = -4$$

$$\therefore b = -2$$

Thus

$$2a = -4 - b$$
$$= -4 + 2$$
$$= -2$$

$$a = -1$$

$\therefore$

$$\begin{matrix} a = -1 \\ b = -2 \end{matrix}$$

5. The polynomial equation $4x^3 - 12x^2 + 5x + 6 = 0$ has roots α, β , and γ . It is known that one of the roots is the sum of the other two. Find α, β , and γ .

3

$$4x^3 - 12x^2 + 5x + 6 = 0$$

$$\begin{cases} \alpha + \beta + \gamma = 3 & \text{--- (1)} \\ \alpha\beta + \alpha\gamma + \beta\gamma = \frac{5}{4} & \text{--- (2)} \\ \alpha\beta\gamma = -\frac{3}{2} & \text{--- (3)} \end{cases}$$

Let $\alpha = \beta + \gamma$

Sub into (1) $\rightarrow 2\alpha = 3$

$$\therefore \alpha = \frac{3}{2}$$

Sub this into (3) $\rightarrow \frac{3}{2} \cdot \beta\gamma = -\frac{3}{2}$

$$\therefore \beta\gamma = -1 \quad \text{--- (4)}$$

Also, $\beta + \gamma = \frac{3}{2} \quad \text{--- (5)}$

From (5), $\gamma = \frac{3}{2} - \beta$

Sub into (4) $\rightarrow \beta(\frac{3}{2} - \beta) = -1$

$$\therefore -\beta^2 + \frac{3}{2}\beta = -1$$

$$\therefore \beta^2 - \frac{3}{2}\beta - 1 = 0$$

$$2\beta^2 - 3\beta - 2 = 0$$

$$(2\beta + 1)(\beta - 2) = 0$$

$$\therefore \beta = -\frac{1}{2} \text{ or } 2$$

(ie $\gamma = 2 \text{ or } -\frac{1}{2}$)

So roots are

$$\frac{3}{2}, 2, -\frac{1}{2}$$

Section B (12 marks)

1. What is the value of $\cos^{-1}(-\frac{1}{2})$? $\cos^{-1}(-\frac{1}{2}) = \pi - \cos^{-1}(\frac{1}{2}) = \pi - \frac{\pi}{3}$ 1

(A) $\frac{\pi}{3}$

(B) $-\frac{\pi}{3}$

(C) $\frac{2\pi}{3}$

(D) $\frac{4\pi}{3}$

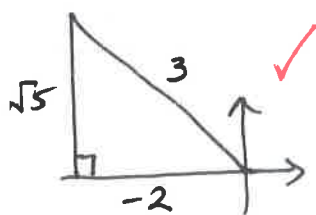
$\therefore \textcircled{C}$ ✓

2. Find the exact values of the following:

(i) $\tan^{-1}(-\sqrt{3}) = -\tan^{-1}(\sqrt{3}) = \textcircled{-\frac{\pi}{3}}$ ✓ 1

(ii) $\sin^{-1}(\sin \frac{7\pi}{5}) = \sin^{-1}[\sin(\pi - \frac{7\pi}{5})]$ 1
 $= \sin^{-1}[\sin(-\frac{2\pi}{5})]$
 $= \textcircled{-\frac{2\pi}{5}}$ ✓

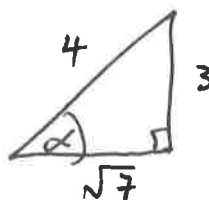
(iii) $\sin(\cos^{-1} \frac{-2}{3})$ 2



$\sin[\cos^{-1} \frac{-2}{3}] = \textcircled{\frac{\sqrt{5}}{3}}$ ✓

(iv) $\tan(2 \sin^{-1} \frac{3}{4})$ 2

$\tan[2 \sin^{-1} \frac{3}{4}]$
 $= \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$
 $= \frac{2(\frac{3}{\sqrt{7}})}{1 - \frac{9}{7}} = \frac{\frac{6}{\sqrt{7}}}{-\frac{2}{7}} = \textcircled{-3\sqrt{7}}$ ✓✓



3.

(i) State the domain and range of the function $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$

2

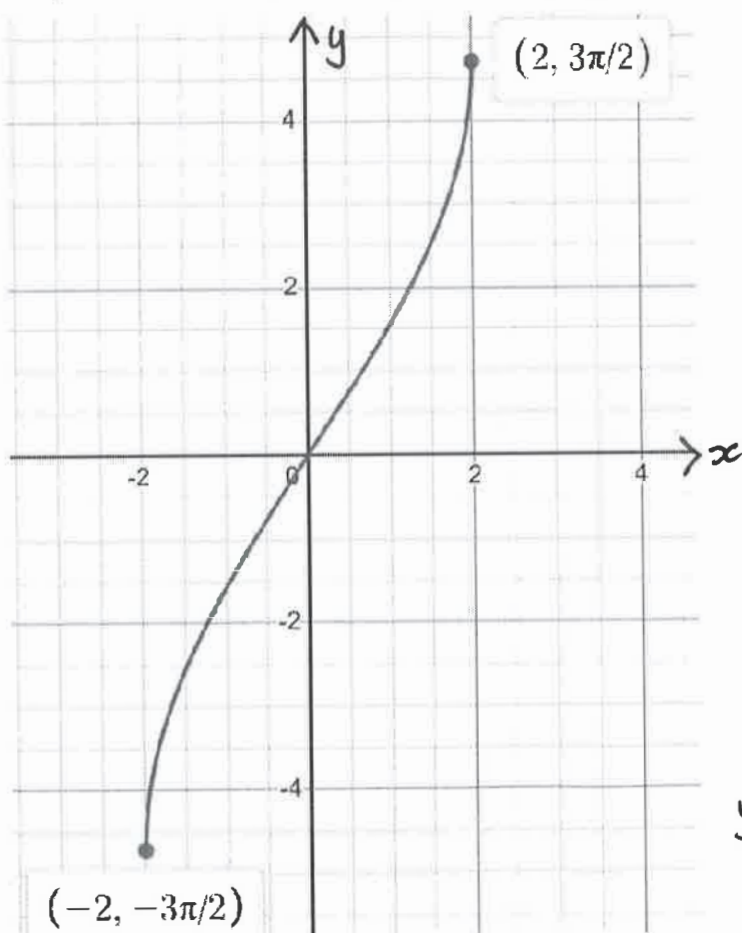
$$-1 \leq \frac{x}{2} \leq 1 \quad \therefore -2 \leq x \leq 2$$

So $D: -2 \leq x \leq 2$ ✓

& $R: -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ ✓

(ii) Sketch the graph $y = f(x)$

2



* label axes

* vertical at end points

* oblique thru (0,0) -



$$y = 3 \sin^{-1}\left(\frac{x}{2}\right)$$

4.

Which diagram represents the domain of the function $f(x) = \cos^{-1}\left(\frac{3}{x}\right)$?

1

(A)



(B)



(C)

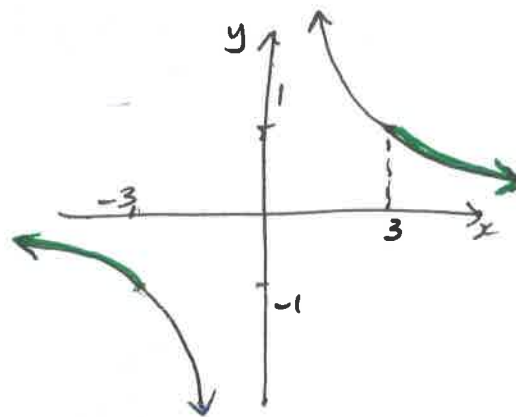


(D)



we need

$$-1 \leq \frac{3}{x} \leq 1$$



So $x \leq -3$ or $x \geq 3$.

$\therefore$ (A) ✓

Section C (9 marks)

1 Find the coefficient of x^2 in the expansion of $(x^2 + \frac{2}{x})^7$

2

The general term is $\binom{7}{k} (x^2)^{7-k} \left(\frac{2}{x}\right)^k$

$$= \binom{7}{k} x^{14-3k} 2^k$$

$\therefore$ we need $14-3k = 2$

$$\therefore \boxed{k=4}$$

$$\text{Thus coefficient of } x^2 = \binom{7}{4} \cdot 2^4$$

$$= \boxed{560}$$

2. Consider the expansion of $(1 + 2x)^n$

(i) Write down an expression for the coefficient of the term in x^4

2

$$(1 + 2x)^n = \sum_{k=0}^n \binom{n}{k} 2^k x^k$$

$\therefore$ coefficient of x^4 term is

$$\binom{n}{4} \cdot 2^4$$

✓✓
(-1 if gives $\binom{n}{4} \cdot 2^4 x^4$)

(ii) The ratio of the coefficient of the term in x^4 to that of the term in x^6 is 5 : 8

Find the value of n .

3

Likewise, coefficient of x^6 term

$$\text{is } \binom{n}{6} 2^6$$

Thus ratio of x^4 term to x^6 term is

$$\frac{\binom{n}{4} \cdot 2^4}{\binom{n}{6} \cdot 2^6} = \frac{\frac{n!}{4!(n-4)!} \cdot 2^4}{\frac{n!}{6!(n-6)!} \cdot 2^6}$$

$$= \frac{30}{4(n-4)(n-5)} = \frac{15}{2(n-4)(n-5)}$$

$$\text{Thus } \frac{15}{2(n-4)(n-5)} = \frac{5}{8}$$

$$\therefore (n-4)(n-5) = 12$$

$$n^2 - 9n + 8 = 0$$

$$\therefore (n-8)(n-1) = 0$$

$$\therefore n = 8$$

(as $n=1$ produces no x^4 or x^6 terms)

3. Consider the binomial expansion $(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$

Show that $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} = n2^{n-1}$

2

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

Differentiate both sides with respect to x :-

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + \dots + n\binom{n}{n}x^{n-1} \quad \checkmark$$

Substitute $x=1$:-

$$n(1+1)^{n-1} = \binom{n}{1} + 2\binom{n}{2} + \dots + n\binom{n}{n}$$

$$\text{i.e. } n \cdot 2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} \quad \checkmark$$

as required #

Section D (10 marks)

1. What is the value of $\sin 2\theta$ given that $\sin \theta = \frac{\sqrt{3}}{4}$ and θ is obtuse?

1

(A) $-\frac{\sqrt{39}}{8}$

(B) $-\frac{\sqrt{3}}{2}$

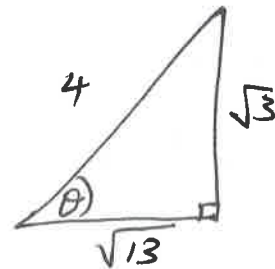
(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{\sqrt{39}}{8}$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \cdot \frac{\sqrt{3}}{4} \cdot -\frac{\sqrt{13}}{4}$$

$$= -\frac{\sqrt{39}}{8} \therefore \text{(A)} \quad \checkmark$$



2. Using the t -substitutions, or otherwise, prove the identity

3

$$\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$$

Let $t = \tan \theta$

$$\therefore \text{LHS} = \frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta}$$

$$= \frac{\frac{2t}{1-t^2} - t}{\frac{2t}{1-t^2} + \frac{1}{t}} \quad \checkmark$$

$$= \frac{\frac{2t - t + t^2}{1-t^2}}{\frac{2t^2 + 1 - t^2}{t(1-t^2)}} \quad \checkmark$$

$$= \frac{t^2(1+t^2)}{t^2+1} \quad \checkmark = t^2 = \tan^2 \theta = \text{RHS.}$$

3. Using an appropriate products-to-sums relationship, or otherwise, solve the equation

$$\sin 5x + \sin x = 0 \quad \text{in the domain } 0 \leq x \leq \frac{\pi}{2}.$$

2

$$\sin 5x + \sin x = 0$$

$$\therefore \sin(3x+2x) + \sin(3x-2x) = 0$$

$$\therefore 2 \sin 3x \cos 2x = 0 \quad \checkmark$$

$$\therefore \sin 3x = 0$$

or

$$\cos 2x = 0$$

$$3x = 0, \pi, 2\pi, 3\pi, \dots$$

$$x = 0, \frac{\pi}{3}$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$x = \frac{\pi}{4}$$

$$\therefore x = 0, \frac{\pi}{3} \text{ or } \frac{\pi}{4} \quad \checkmark$$

4. The quadratic equation $ax^2 + bx + c = 0$ has roots $x = \tan \alpha$ and $x = \tan \beta$.

(i) Show that $\tan(\alpha + \beta) = \frac{b}{c-a}$

2

$$\begin{aligned} \text{Sum of roots} &= \tan \alpha + \tan \beta = -\frac{b}{a} \\ \text{product of roots} &= \tan \alpha \tan \beta = \frac{c}{a} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Sum of roots} \\ \text{product of roots} \end{aligned}} \right\} \checkmark$$

$$\text{So } \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{-\frac{b}{a}}{1 - \frac{c}{a}} = \frac{-b}{a-c} = \frac{b}{c-a} \quad \checkmark$$

as required.

(ii) Show that $\tan^2(\alpha - \beta) = \frac{b^2 - 4ac}{(a+c)^2}$

2

$$\tan^2(\alpha - \beta) = \left(\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \right)^2$$

$$= \frac{\tan^2 \alpha - 2 \tan \alpha \tan \beta + \tan^2 \beta}{(1 + \tan \alpha \tan \beta)^2}$$

$$= \frac{(\tan \alpha + \tan \beta)^2 - 4 \tan \alpha \tan \beta}{(1 + \tan \alpha \tan \beta)^2} \quad \checkmark$$

$$= \frac{\left(\frac{b}{a}\right)^2 - \frac{4c}{a}}{\left(1 + \frac{c}{a}\right)^2}$$

$$= \frac{b^2 - 4ac}{(a+c)^2} \quad \checkmark$$

Section E (12 marks)

1. Solve the inequality $\frac{2x}{1-x} \geq x$ ($x \neq 1$)

3

$$\frac{2x}{1-x} \geq x$$

$$\therefore 2x(1-x) \geq x(1-x)^2 \quad \checkmark$$

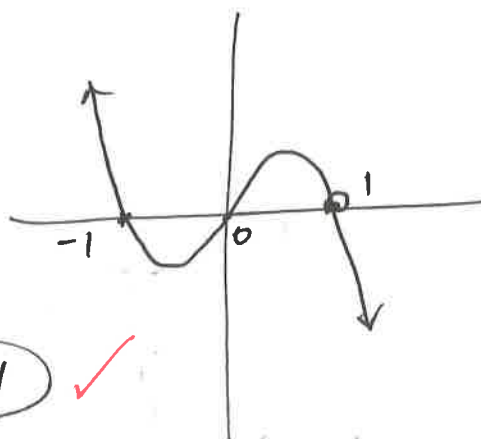
$$\therefore 2x(1-x) - x(1-x)^2 \geq 0$$

$$x(1-x) [2 - (1-x)] \geq 0$$

$$x(1-x)(1+x) \geq 0 \quad \checkmark$$

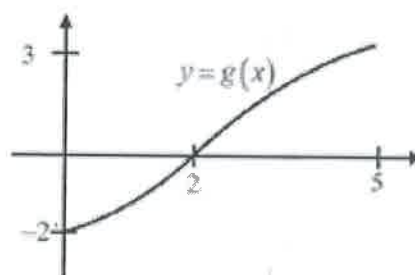
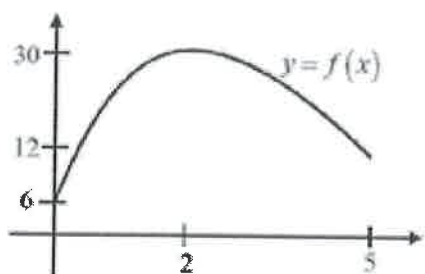
$$-x(x-1)(x+1) \geq 0$$

$$\therefore x \leq -1 \text{ or } 0 \leq x < 1 \quad \checkmark$$



2. Consider the following sketches of the functions $y = f(x)$ and $y = g(x)$.

1



What is the range of the function $y = \frac{f(x)}{g(x)}$? [Circle the answer]

(A) $(-\infty, -3] \cup [4, \infty)$

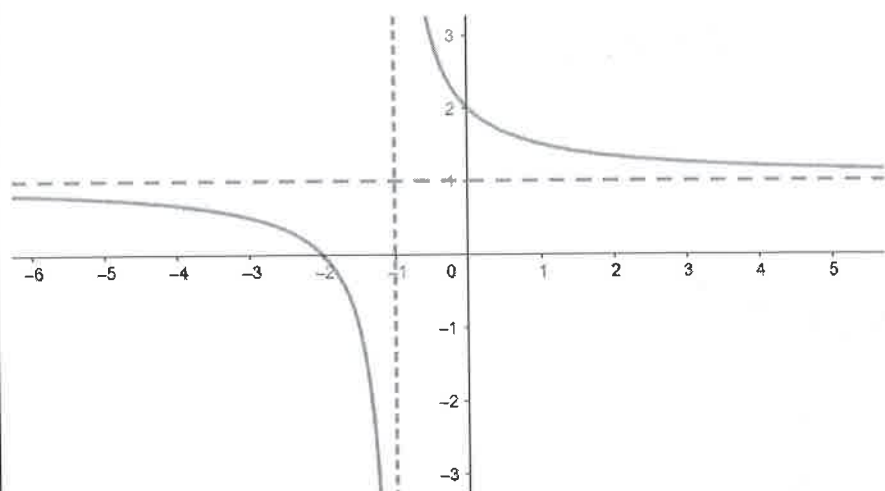
(B) $[-3, 4]$

(C) $[-15, 2]$

(D) all real y

answer: (A) $\checkmark$

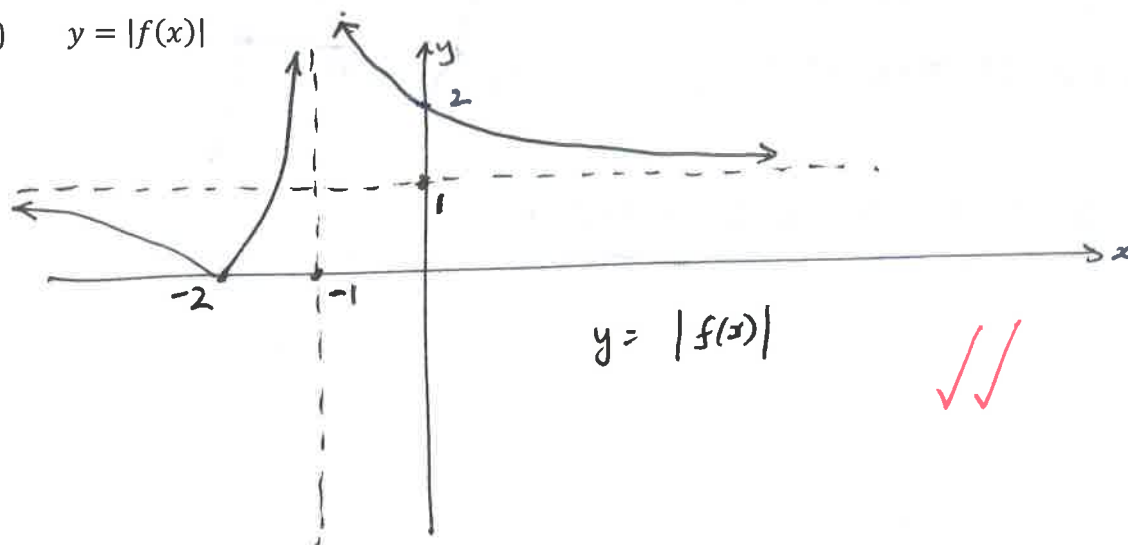
3. The graph of $y = f(x)$ is shown below:



Draw separate one-third page sketches of the following curves, clearly indicating any important features such as asymptotes, intercepts and turning points.

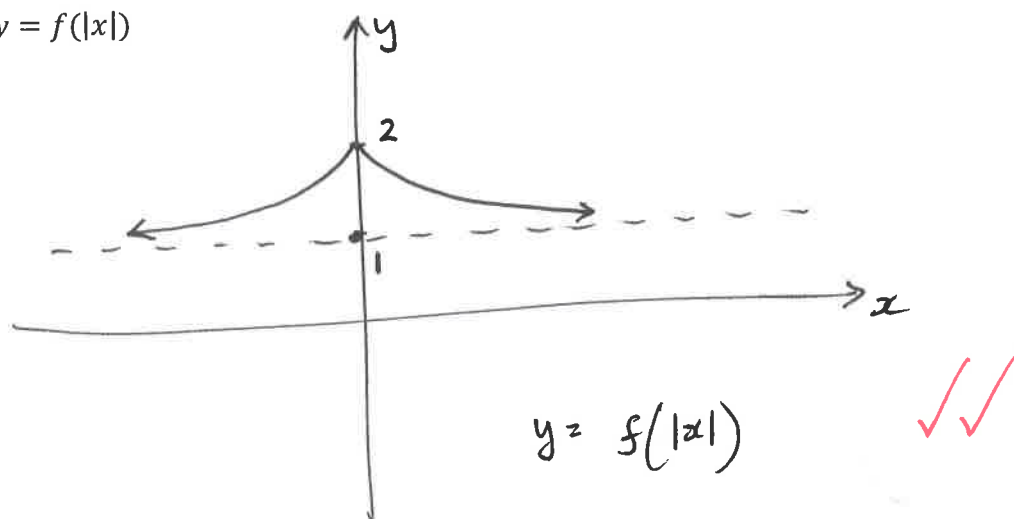
(i) $y = |f(x)|$

2

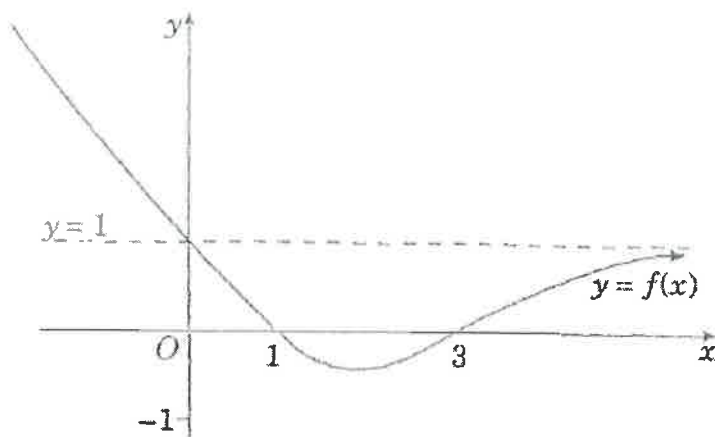


(ii) $y = f(|x|)$

2

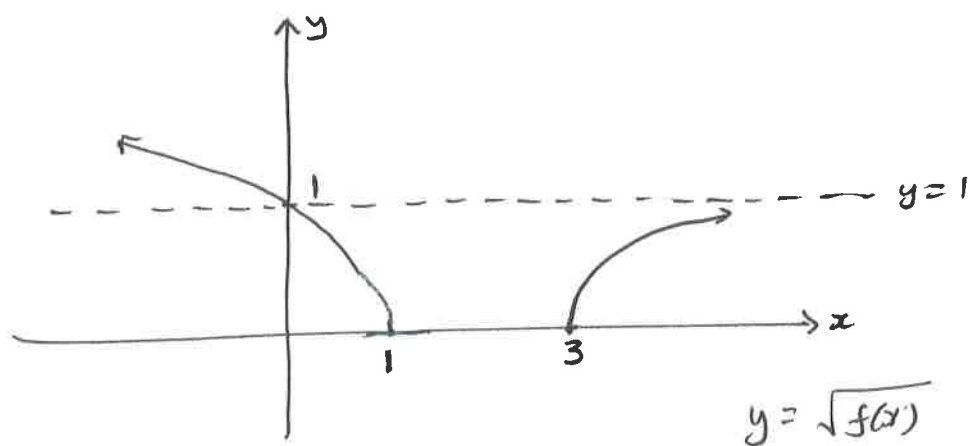


4. The graph shows the graph of the function $y = f(x)$. It has a horizontal asymptote $y = 1$.



Draw a one-third page graph of $y = \sqrt{f(x)}$ showing any asymptotes and stating its domain and range.

3

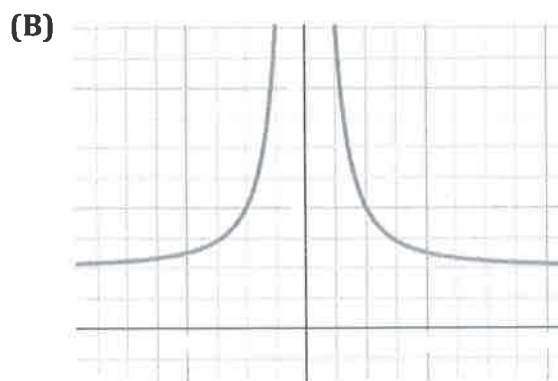
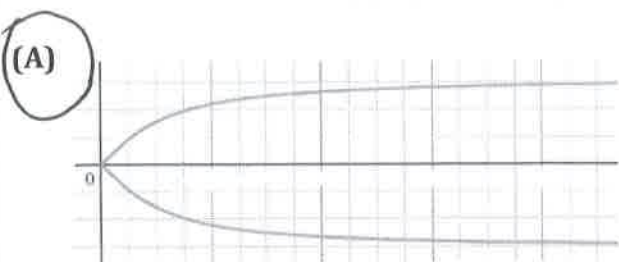


$$D: x \leq 1 \text{ or } x \geq 3$$

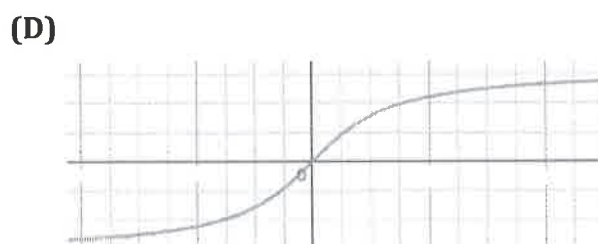
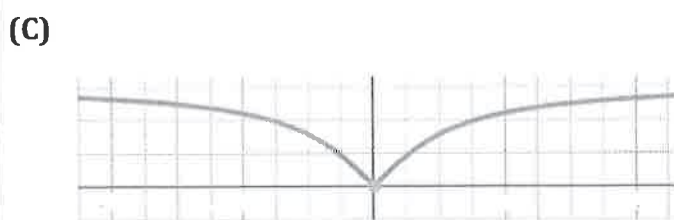
$$R: y \geq 0.$$

5. Which of the following graphs best represents $|y| = \tan^{-1}(x)$?

1



ANSWER: (A) ✓



Section F (8 marks)

1. The surface area of a sphere of radius r metres is decreasing at a rate of 0.8 m^2 per second at the instant when $r = 2.3 \text{ m}$.

Calculate the rate of decrease of the radius of the sphere at this instant.

2

$$S = 4\pi r^2 \therefore \frac{ds}{dr} = 8\pi r \quad \frac{ds}{dt} = 0.8 \text{ m}^2/\text{s}$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{dr}{ds} \cdot \frac{ds}{dt} \\ &= \frac{1}{8\pi r} \cdot (0.8) \end{aligned} \quad \checkmark$$

$\therefore$ When $r = 2.3$,

$$\begin{aligned} \frac{dr}{dt} &= \frac{0.8}{8\pi(2.3)} = \frac{1}{23\pi} \text{ m/s} \\ &\approx 0.0138396 \text{ m/s} \end{aligned} \quad \checkmark$$

2. A metal statuette is taken from a freezer at -8°C into a room where the air temperature is 22°C .

The rate at which the statuette warms follows Newton's law, that is $\frac{dT}{dt} = -k(T - 22)$ where k is a positive constant, time t is measured in minutes, and temperature T is measured in degrees Celsius.

- (i) Show that $T = 22 - Ae^{-kt}$ is a solution of the equation $\frac{dT}{dt} = -k(T - 22)$, and find the value of A . 2

$$\begin{aligned}\frac{dT}{dt} &= (-k)(-Ae^{-kt}) \\ &= kAe^{-kt} \\ &= k(22 - T) \\ &= -k(T - 22)\end{aligned}$$

$\therefore$ is solution. ✓

when $t=0$, $T=-8$

$$\therefore -8 = 22 - A$$

$$\therefore A = 30 \quad \checkmark$$

- (ii) The temperature of the statuette reaches 4°C in 90 minutes. Find the exact value of k . 2

when $t=90$, $T=4$

$$\therefore 4 = 22 - 30e^{-90k} \quad \checkmark$$

$$30e^{-90k} = 18$$

$$e^{-90k} = \frac{18}{30} = \frac{3}{5}$$

$$\therefore -90k = \ln \frac{3}{5}$$

$$\therefore k = -\frac{1}{90} \ln \left(\frac{3}{5} \right) \quad \checkmark$$

$$\left(k = \frac{\ln 5 - \ln 3}{90} \right)$$

- (iii) Find the temperature of the statuette after another 90 minutes. 1

when $t=180$, $-180k$

$$T = 22 - 30e^{-180 \left[-\frac{1}{90} \ln \frac{3}{5} \right]}$$

$$= 22 - 30e$$

$$= 22 - 30e^{2 \ln \frac{3}{5}} = 22 - 30 \times \frac{9}{25} = 22 - \frac{54}{5}$$

$$T = 11.2^{\circ}\text{C} \quad \checkmark$$

3. When added to water, 5 grams of a substance dissolves at a rate equal to 10% of the amount of undissolved chemical per hour. If x is the number of grams of *undissolved* chemical after t hours, which equation describes the rate of change of x ? 1

(A) $\frac{dx}{dt} = -\frac{1}{10}x$

(B) $\frac{dx}{dt} = -\frac{1}{5}x$

(C) $\frac{dx}{dt} = \frac{1}{5}(10 - x)$

(D) $\frac{dx}{dt} = \frac{1}{10}(5 - x)$

answer :

(A) ✓

END OF EXAM